

List

$L = [1, 2, 3]$

L

1 | 2 | 3

0 1 2

$L = LL$

$\text{print}(\text{id}(L))$

$L1 = L$

$L1$

1 | 2 | 3

1 | 2 | 3

Shallow Copy

Deep Copy

$L1 = L.\text{copy}()$

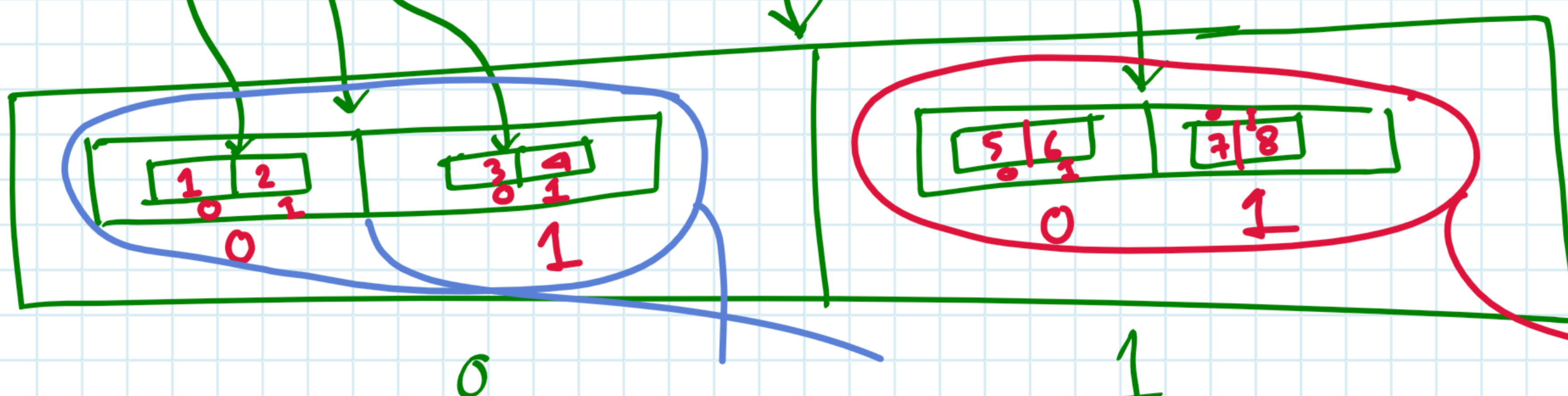
$L1 = L$

$L = [1, 2, 3]$

$L1 = L$

$L1 \rightarrow$

$\text{list_demo} \rightarrow [[1, 2], [3, 4], [5, 6], [7, 8]]$



$\text{print}(\text{list_demo}[0]) \rightarrow [1, 2], [3, 4]$

$\text{print}(\text{list_demo}[1]) \rightarrow [5, 6], [7, 8]$

$\text{print}(\text{list_demo}[0][1]) \rightarrow$

$\text{list_demo}[1][0:2] \rightarrow [5, 6], [7, 8]$

Heterogenous

$L = [1, \text{True}, 5+6j, 5.5, \text{"hello"}]$

$\text{type}(L[2]) \rightarrow \text{Complex}$

$L1 \rightarrow \text{list}(\text{"hello"}) \rightarrow \text{h | e | l | l | o}$

$\text{print}(L1) \rightarrow ['h', 'e', 'l', 'l', 'o']$

Accessing items from a list

positive index → 0 1 2 3 4 5
 $L = [1, 2, 3, 4, 5, 6]$
negative index → -6 -5 -4 -3 -2 -1

Slicing → $L[start : stop : step]$

$L[1:5] \rightarrow [2, 3, 4, 5]$

$L[1:5:+2] \rightarrow [2, 4]$

$L[-1:-5:+1] \rightarrow$

$L[::-1] \rightarrow [6, 5, 4, 3, 2, 1]$

Adding items

Append

$L = [1, 2, 3, 4, 5]$

$L.append(6)$

$[1, 2, 3, 4, 5, 6]$

$L.append([6, 7, 8])$

$[1, 2, 3, 4, 5, [6, 7, 8]]$

extend

$L.extend([6, 7, 8])$
List

$[1, 2, 3, 4, 5, 6, 7, 8]$

$L.extend('Dhaka')$

$[1, 2, 3, 4, 5, 'D', 'h', 'a', 'k', 'a']$

Insert

$L = [1, 2, 3, 4, 5]$

$L.insert(1, 100)$

$[1, 100, 2, 3, 4, 5]$

Editing a List

$L = [1, 2, 3, 4, 5]$

$L[0] = 6$

$[6, 2, 3, 4, 5]$

$L[-1] = 500$

$[6, 2, 3, 4, 500]$

$L[1:4] = [200, 300, 400]$

$[6, 200, 300, 400, 500]$
1 2 3 4

Delete a List

del L = [1, 2, 3, 4, 5]

del L[-1]

del L[1:3]

pop

L.pop() → [1, 2, 3, 4]

remove

L = [1, 2, 3, 4, 5]

L.remove(5)

↑

LIFO Stack

push, pop

Queue
FIFO

→

clear

L.clear()

L → []

Operations in List

L1 = [1, 2, 3]

L2 = [5, 6, 7, 8]

print (L1 + L2) // Concatenation/Merge

[1, 2, 3, 5, 6, 7, 8]

print (L1 * 3)

→ [1, 2, 3, 1, 2, 3, 1, 2, 3]

[1, 2, 3] [1, 2, 3] [1, 2, 3]

Membership operators

5 not in L1 → T/F

5 in L1 → T/F

L2 → [1, 2, 3, 4, [5, 6]]

if [5, 6] in L2 :

print ("Got the list")

else :

print ("Nai vai ekhane")

len(L1)

Loops

L1 = [1, 2, 3, 4, 5]

for i in L1:

print(i)

1
2
3
4
5

for i in range(len(L1)):

print(i)

`for i in range(len(L1)):`
`print(L1[i])`

0 1 2 3 4 → 5

$L1 \rightarrow [1, 2, 3, 4, 5]$
0 1 2 3 4

`for i in L1:`
`print(i)`

$L = [\underbrace{[1, 2], [3, 4]}_0, \underbrace{[5, 6], [7, 8]}_1]$

`for i in (L):`
`print(i)`

$[[1, 2], [3, 4]]$
 $[5, 6], [7, 8]$

functions $L = [1, 2, 3, 4, 5]$
→ `len(L) → 5`

`min(L) → 1`

`max(L) → 5`

`sorted(L) → [1, 2, 3, 4, 5]` ↗ new list → sorted

`sorted(L, reverse=True) → [5, 4, 3, 2, 1]`

`L.count(5) → 1`
↑
value

`L.index(1) → 0`

Difference?

`L.reverse()`

`[5, 4, 3, 2, 1]`

`L.sort()`

`[1, 2, 3, 4, 5]`

→ existing list sorted

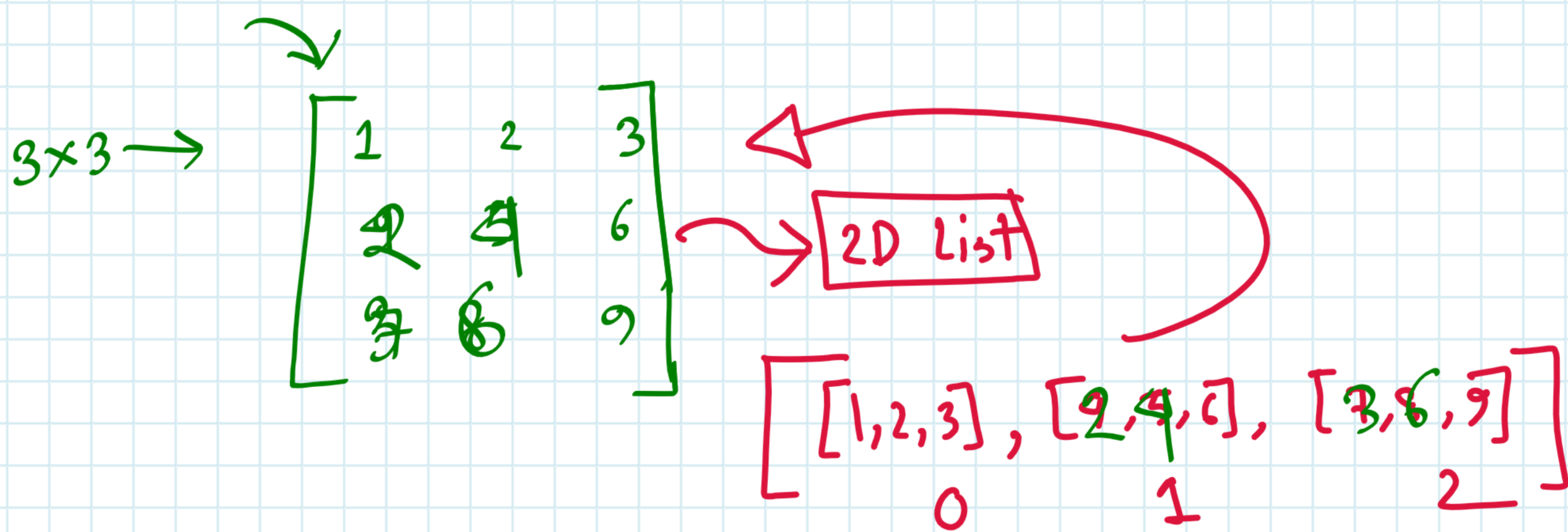
→ list comprehension
→ zip

List Comprehension

```
L = []  
for i in range(1, 11):  
    L.append(i)  
  
print(L)
```

~~New~~list = [expression for item in iterable if cond == True]

✓ [1, 50] → print/store in a list → divisible by 5



matrix = []

```
for j in range(1, 4):  
    row = []  
    for i in range(1, 4):  
        row.append(i * j)  
    matrix.append(row)  
  
print(matrix)
```

j → 1, 2, 3

j = 1

i = [1, 2, 3]

j = 2

row = [1, 2, 3]

i = [1, 2, 3]

row = [2, 4, 6]

j = 3

matrix = [[1, 2, 3], [2, 4, 6], [3, 6, 9]]
0

[i * j for i in range(1, 4)]

j = 1

[1, 2, 3]

L = [i for i in range(1, 4)]

[1, 2, 3]
0 1 2

L = [[i] for i in range(1, 4)]
[[1]] [[2]] [[3]]

$[[\textcircled{i}] \text{ for } i \text{ in range}(1,4)]$

$[[1,2,3], [2,4,6], [3,6,9]]$ $[[1], [2], [3]]$

↑ ↑ ↑ ↑

3 3 3 3

$[[i*j \text{ for } i \text{ in range}(1,4)] \text{ for } j \text{ in range}(1,4)]$